# Ahmed Shaboury

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*Mechatronics Engineering Student with a solid foundation in Robotics, Embedded Systems, Autonomous Systems and AI*

## Experience

**Valeo - Software Engineering Intern** November 2025 - June 2026

- Contributed to the development of a **BMW Smart Parking System** within Valeo's R&D Center (Brain Division - anSWer team), focusing on software quality and validation.
- Implemented unit tests using **Google Test (GTest)** for multiple critical modules, improving reliability and achieving high test coverage.
- Developed and executed **CAPL scripts** for automated test scenarios and in-vehicle communication validation.
- Applied **static code analysis tools** to ensure compliance with automotive software quality standards and enhance robustness.
- Collaborated in an embedded **C++** development environment, supporting testing workflows, debugging, and integration activities.
- Worked within a professional Git-based workflow (branching, merging, issue tracking) while coordinating with cross-functional and international teams.
- Contributed to automation efforts by developing scripts to streamline parameter configuration and reduce manual testing overhead.

**STMicroelectronics - System Programming in Linux Intern** September 2025 - February 2026

- Gained hands-on experience in Linux system-level programming using C and C++.
- Worked on filesystem management, process control, and multithreading programming under Linux.
- Built and configured minimal Linux distributions using the Yocto build system.
- Implemented socket-based client-server communication and network programming utilities.

**ITI - Embedded Systems Trainee** June 2025 - August 2025

- Completed intensive embedded systems training program focusing on microcontroller programming, real-time systems, and hardware-software integration
- Gained hands-on experience with AVR ATMega32 microcontrollers, embedded C programming, and communication protocols (UART, SPI, I2C)
- Developed practical skills in circuit analysis, sensor interfacing, and interrupt handling for embedded applications

**NTI - Machine Learning Trainee** June 2025 - August 2025

- Gained hands-on experience in data preprocessing, feature engineering, and building predictive models with supervised and unsupervised learning algorithms.
- Worked on Time Series Analysis for forecasting tasks, applying models such as ARIMA and Prophet to real datasets.
- Utilized Apache Spark for large-scale data processing and distributed machine learning workflows.

## Projects

**Autonomous Warehouse Mobile Robot (Graduation Project)** **June 2025 - June 2026**

Designed and implemented a full-stack autonomous mobile robot system for warehouse environments using ROS2 Humble on **NVIDIA Jetson Nano** and Arduino. Developed a modular software architecture integrating navigation, localization, and control subsystems. Implemented autonomous navigation using ROS2 Nav2 stack with global and local planners, costmaps, and recovery behaviors. Built SLAM-based mapping and localization using SLAM Toolbox for real-time operation in unknown indoor environments. Integrated multiple sensors and developed system-level data fusion pipelines to enhance robustness and navigation accuracy. Contributed to end-to-end system integration, debugging, and optimization in a Linux-based embedded environment using C++ and Python ROS2 nodes.

**ROS2 Launch Tester** **December 2024**

Developed an automated launch file validation tool for ROS2 systems that addresses silent failures in complex multi-node deployments. The tool uses YAML configuration files to validate expected nodes, topics, and services, reducing debugging time by 60% and enabling reliable CI/CD integration. Features auto-discovery mode, configurable timeouts, color-coded reporting, and smart waiting strategies using ROS2's introspection capabilities via rclpy. GitHub Repository

**Automotive Wiper System Controller (Model-Based Development)** **August 2025**

Designed and implemented a MATLAB/Simulink automotive wiper controller with multi-mode operation (Off, Low, High, Auto) and adaptive PWM control mapped across 8 speed levels. Achieved 10 ms execution cycle using a 100 Hz scheduler for real-time compliance. Integrated safety mechanisms, fail-safe logic, and smooth motor transitions. Enabled automatic C code generation and validated performance through simulation.

**F1Tenth Autonomous Racing Controller** **February 2024 - August 2024**

Ranked 1st locally and 11th globally in F1TENTH Autonomous Racing competition by implementing a Wall-Follower PID controller. Developed features including PID control for smooth following, dynamic speed adjustment, safety collision prevention measures, and ROS2 integration for AutoDrive simulator communication.

## Education

**Bachelor's in Mechatronics & Robotics Engineering** **Mansoura University** **2022 - 2026**
Focus: Autonomous & Robotics Systems **GPA: 3.92/4.0**

## Skills

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|---|---|
| **Programming Languages** | C/C++, Python, MATLAB, Java, Assembly, SQL, PLC |
| **Robotics & Frameworks** | ROS & ROS2, OpenCV, NumPy, Pandas |
| **Development Tools** | Git, Docker, SIMULINK, Gazebo, CARLA Simulator, Coppelia Sim |
| **Languages** | English (C1), Arabic (Native), German (A2) |

## Achievements

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|---|---|
| **F1Tenth Competition** | Ranked 1st Locally and 11th Globally in Autonomous Racing Competition |
| **Academic Excellence** | Top 3 student in Mechatronics & Robotics Engineering at Mansoura University |
| **Current GPA** | 3.92/4.0 in Mechatronics & Robotics Engineering |